Movement and Control of Biomimetic Humanoid Robots

Ben P. Bolen · PhD Defense
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Portland State UNIVERSITY

AGILE & ADAPTIVE ROBOTICS LABORATORY

C³NS Communication, Coordination, and Control in Neuromechanical Systems

NeuroNex

Biology offers design principles for unstructured environments

Prepared ground	Unstructured terrain
Wheels exploit a predictable surface	Legs negotiate changing footholds
Efficiency through simple contact	Compliance can store and return energy
Known operating conditions	Feedback must adapt to disturbances

Research goal: useful autonomy beyond prepared surfaces

Biomimetic robots can also test biological hypotheses



Biomechanics and neuroscience

Controlled changes in mechanics and sensory pathways

Further application: improved prosthetic and orthotic devices

The AARL humanoid is our platform for this research





Skeleton and joints

Artificial muscles

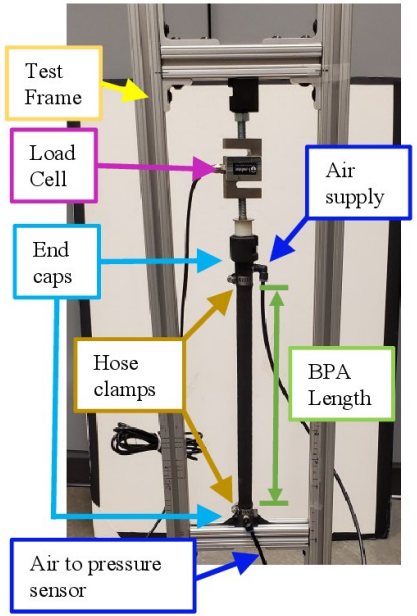
Sensory neural control

Biological inspiration must be checked against measured function

Four knowledge gaps organize this dissertation

Section	Question addressed
1. BPA actuator characterization	How does actuator choice change isometric force?
2. Robot joint design factors	How much force becomes delivered joint torque?
3. SADb	Which sensory evidence informs a pathway choice?
4. SNS tools	How can coupled mechanisms be tested and interpreted?

Actuator selection requires force predictions across resting lengths



Design choice: resting length

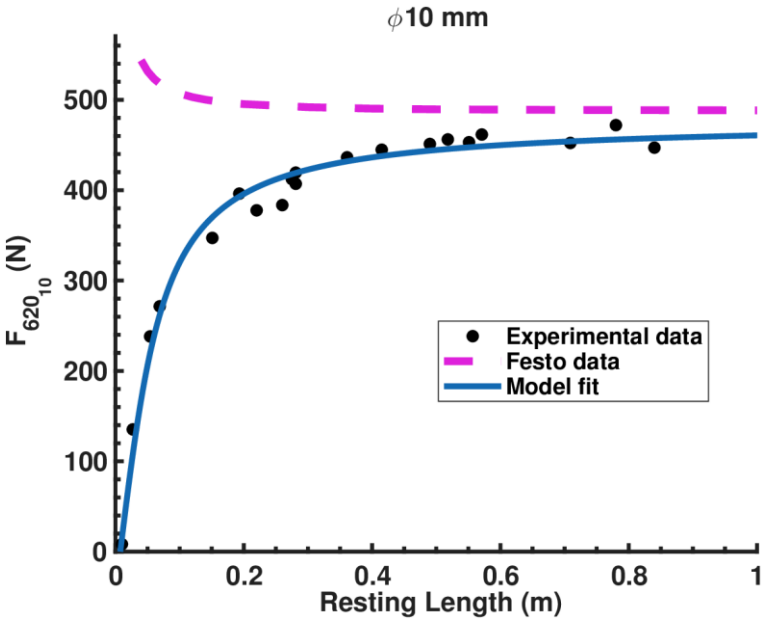
Measurements: length, pressure, contraction, and force

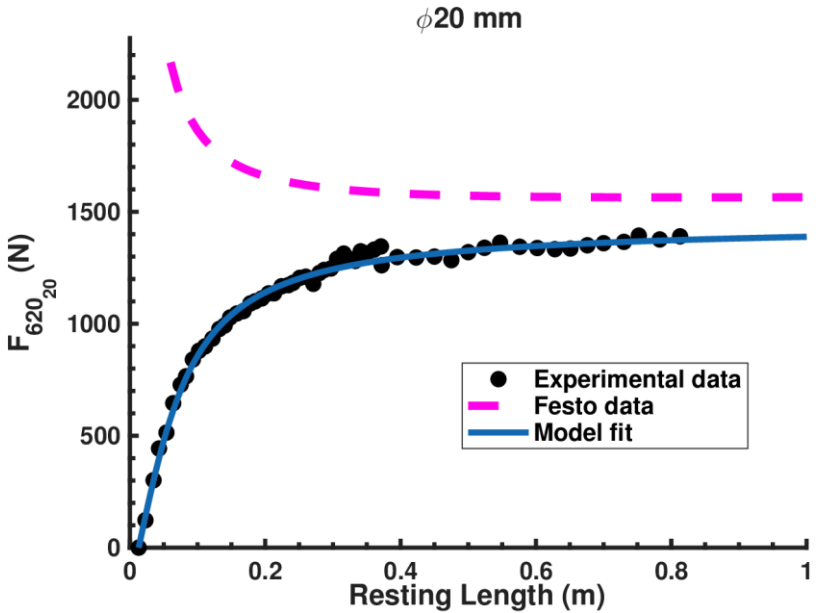
537 isometric observations

Question 1: what force can the selected actuator produce?

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

Shorter BPAs deliver less maximum isometric force



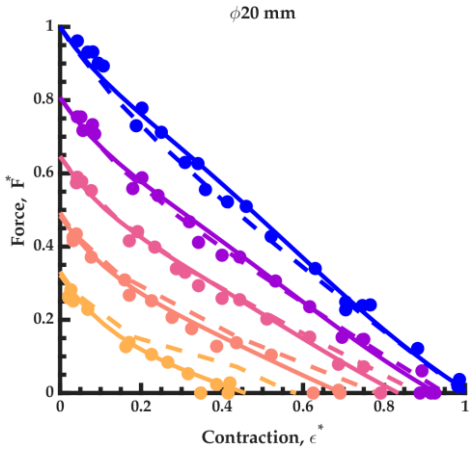


Resting length changes the force available to a robot design

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

Resting-length scaling completes the normalized force model

$$F(\varepsilon^*, P^*, l_0) = F^*(\varepsilon^*, P^*) \cdot F_{620}(l_0)$$



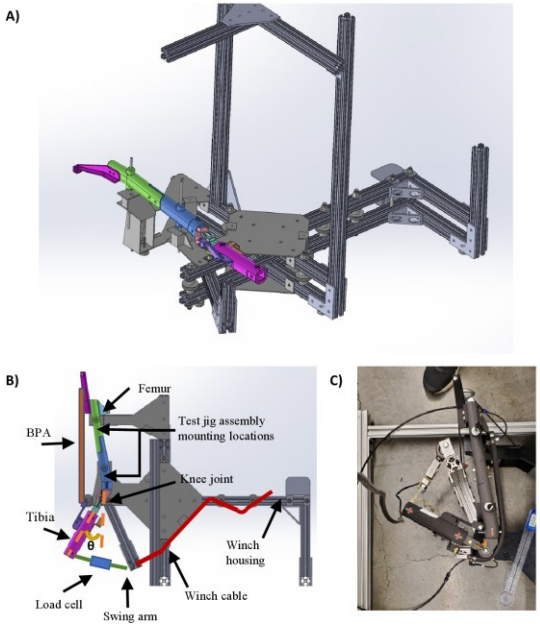
F* normalized force
P* pressure / 620 kPa
ε* normalized contraction
l_o resting length

Curves: 200–620 kPa
Solid model; circles measured
Dashed manufacturer

Diameter-specific coefficients remain fixed from the published study

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

A force model must be carried through the joint transmission



Actuator force

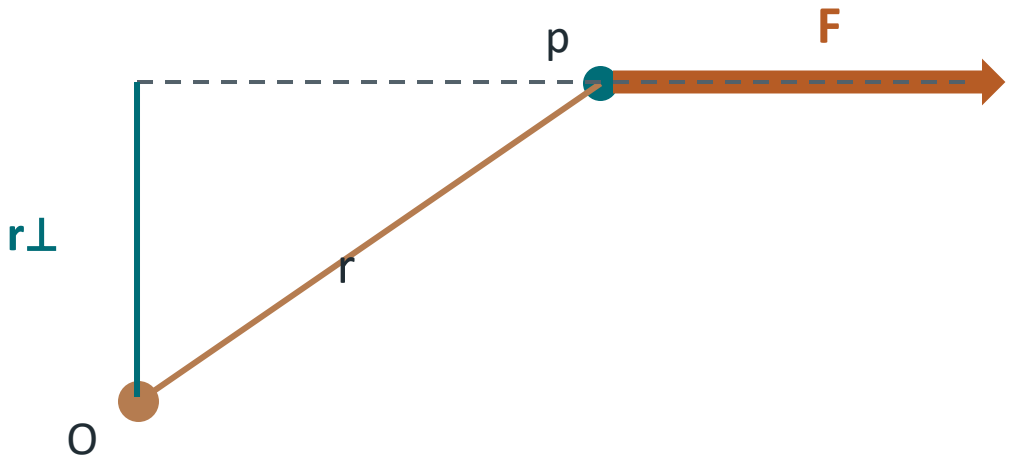
Routing and attachment mechanics

Delivered joint torque

Question 2: how much of the available force becomes useful torque?

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Joint torque depends on force direction and moment arm



$$\tau_z = (r \times F) \cdot \hat{z}$$

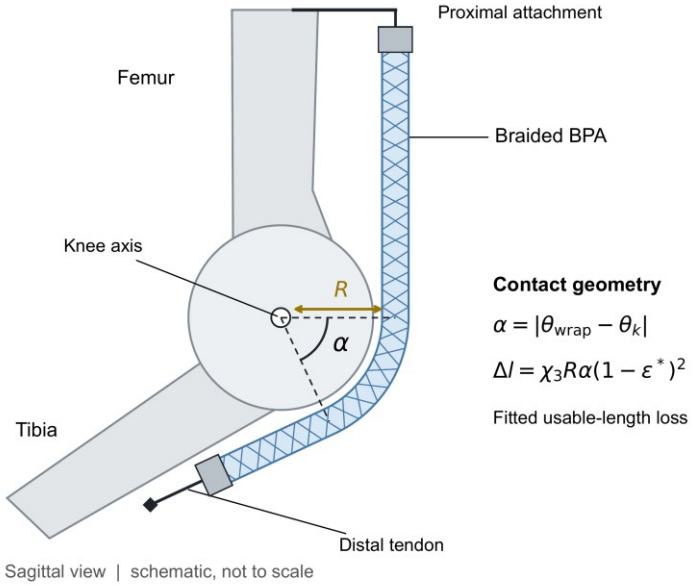
$$|\tau_z| = F\, r_\perp$$

Routing also changes actuator length

A force model needs a transmission model to predict delivered torque

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Length, compliance, and wrapping alter usable contraction

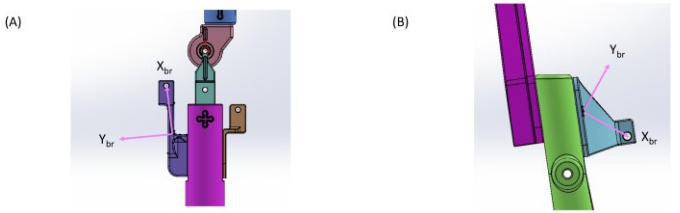


Contact geometry

$\alpha = |\theta_{wrap} - \theta_k|$

$\Delta l = \chi_3 R \alpha (1 - \varepsilon^*)^2$

Fitted usable-length loss



Constant offset

Load-dependent deflection

Contact-dependent length loss

The correction terms belong to the actuator–attachment system

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Corrections reduce error on held-out knee tests

Validation RMSE, N m	No corrections	Corrected
Flexor, 2 tests	7.588	1.490
Extensor, 3 tests	2.554	1.098

Same stiffness pair across both configurations

Tests reserved before evaluating the adopted fit

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

The corrected model guides a buildable two-BPA cam route



Two 20 mm BPAs

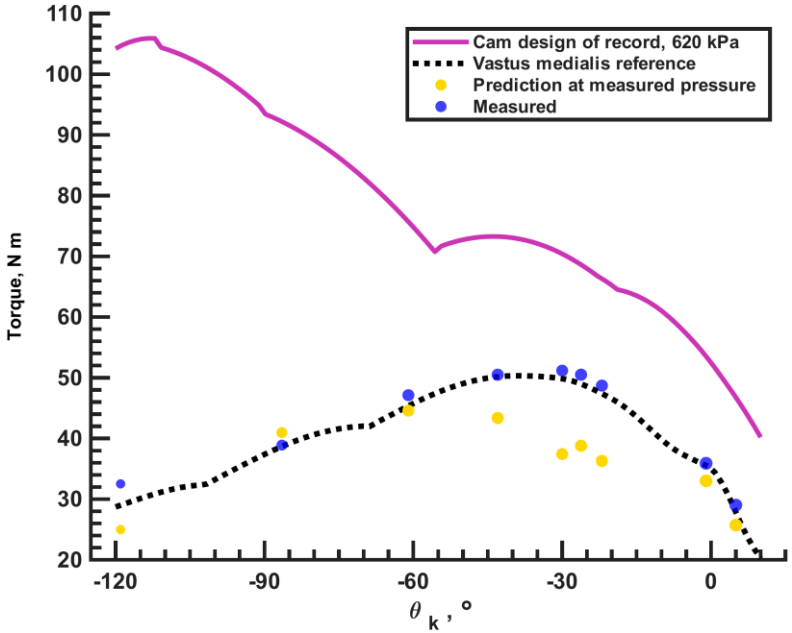
Cam-assisted tendon path

Adopted transmission corrections guide the design

The route must satisfy force, excursion, and geometric constraints

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

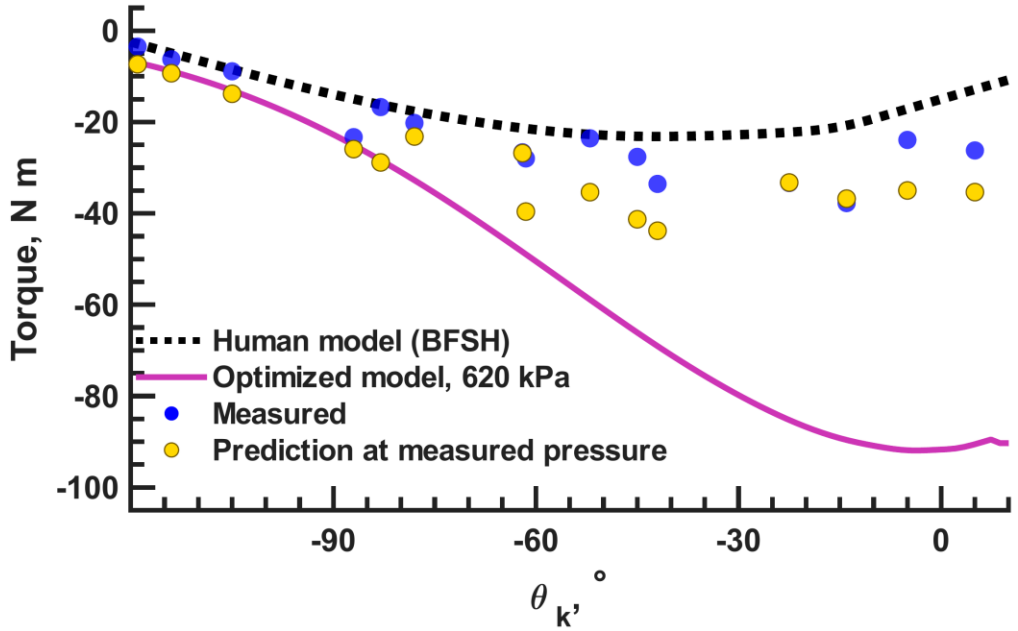
All nine extensor measurements meet the human target



9 of 9 target margins positive · +0.24 to +3.52 N m

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

All fifteen flexor measurements meet the human target



15 of 15 targets met · 226.5–638 kPa · negative torque denotes flexion

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Measured capacity makes feedback design a meaningful next question

Mechanical foundation established	Next question
Resting-length-aware force prediction	Which sensory signals should regulate it?
Corrections for the joint transmission	Under what experimental conditions?
9 extensor + 15 flexor target comparisons	What pathway effect should be tested?

Mechanical capacity and sensory control answer different questions

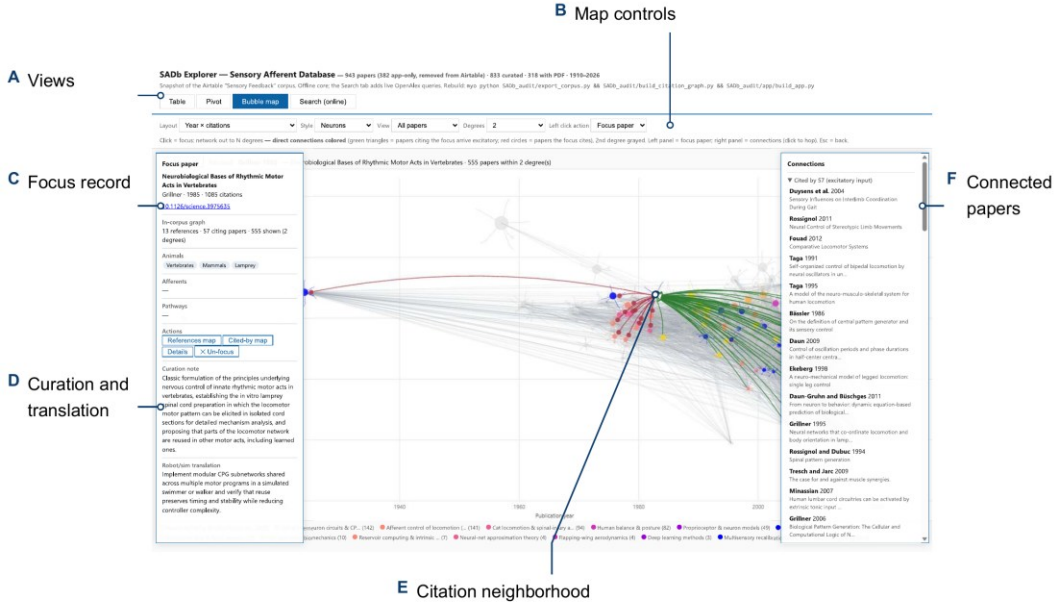
Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Sensory findings are distributed across experimental contexts

A modeling decision needs…	**The record must retain…**
A sensory signal	Receptor and measured quantity
A connection sign	Target pathway and experimental conditions
A task-dependent effect	Species, preparation, and behavioral phase
A defensible assumption	Source paper and translation notes

Dissertation Ch. 5; Appendix E

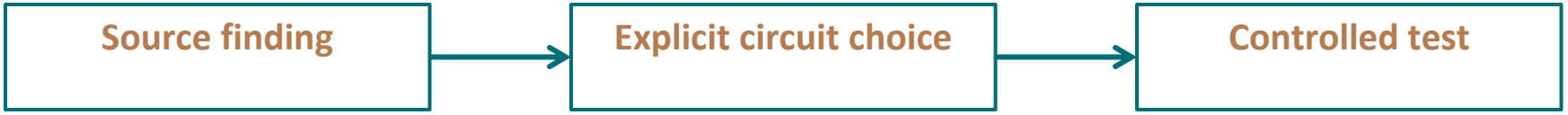
SADb makes sensory evidence traceable to its source context



1,393 total records · search, pivot, and follow citation links

Dissertation Ch. 5, The Resource · Appendix E, distributed app snapshot

Source context turns a candidate pathway into a testable assumption



Signal and experimental context

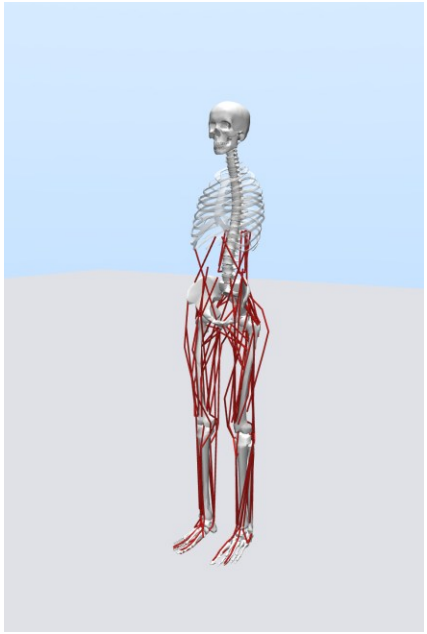
Connection sign, target, and selected parameters

Matched input or pathway intervention

A source-grounded pathway still needs implementation and functional checks

Dissertation Ch. 5, Use in Neural-Controller Design; Ch. 6

The walker attempts exposed a need for interpretable model tests



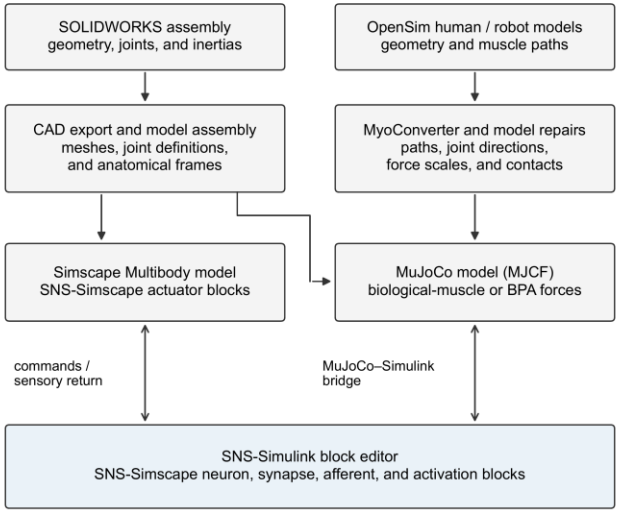
A circuit can oscillate while its mechanical model fails

Implementation, feedback, and mechanics must be separated

Question 4: what caused the observed neural–mechanical behavior?

Dissertation Ch. 6, opening narrative and tools gap

The model workflow supports distinct mechanical implementations



Command exchange, circuit agreement, and motion each require their own check

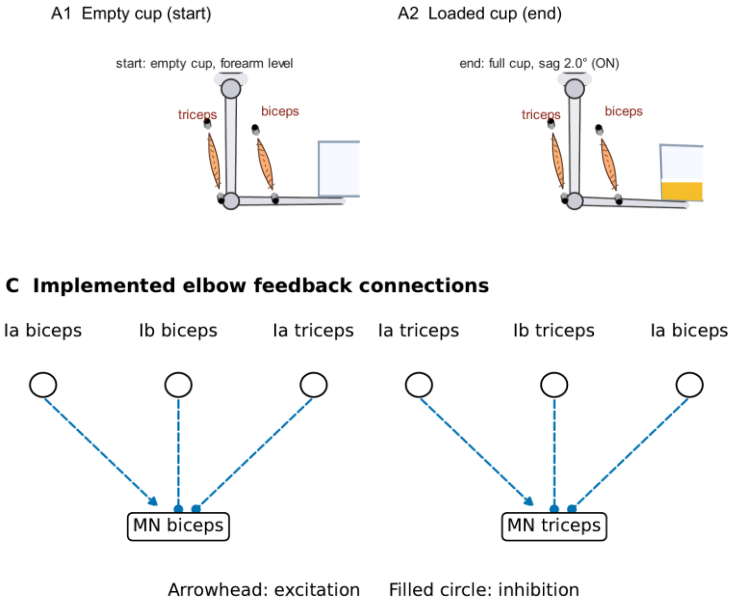
Dissertation Ch. 6, Model construction and neural-controller interfaces

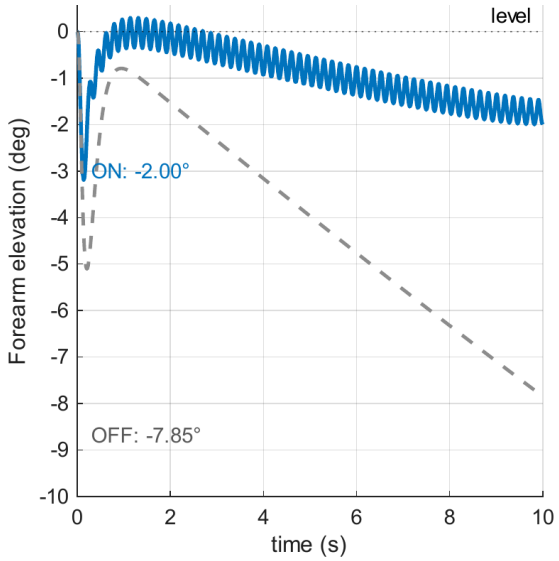
Staged tests separate rhythm, feedback, and mechanical support

Condition introduced	Question tested
Suspended, without sensory feedback	Can the circuit generate activity?
Sensory feedback	How does feedback reshape the output?
Standing and perturbations	Can the model maintain posture?
Supported ground contact	Does physical coordination meet the criterion?
Reference and intervention comparisons	Which mechanism or mismatch explains the result?

Dissertation Ch. 6, curriculum Table 6.2

Sensory feedback reduces sag in the loaded-arm model

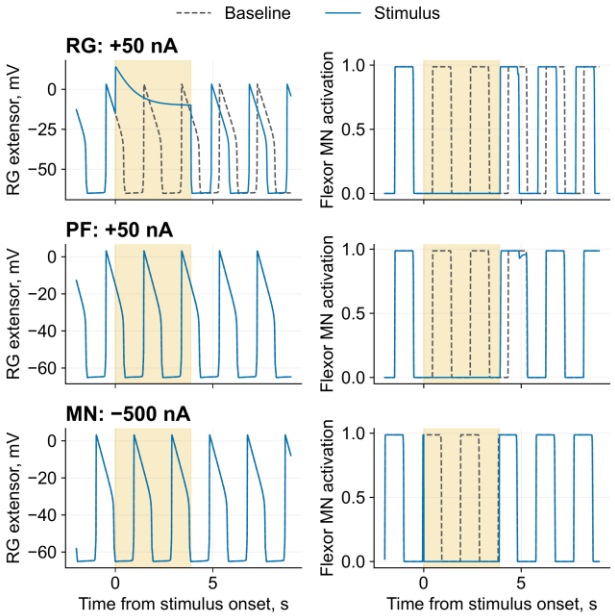




Sag: 2.00° with feedback · 7.85° without feedback

Dissertation Ch. 6 and Appendix D · Current saved verification records

Two deleted bursts can preserve or reset the rhythm



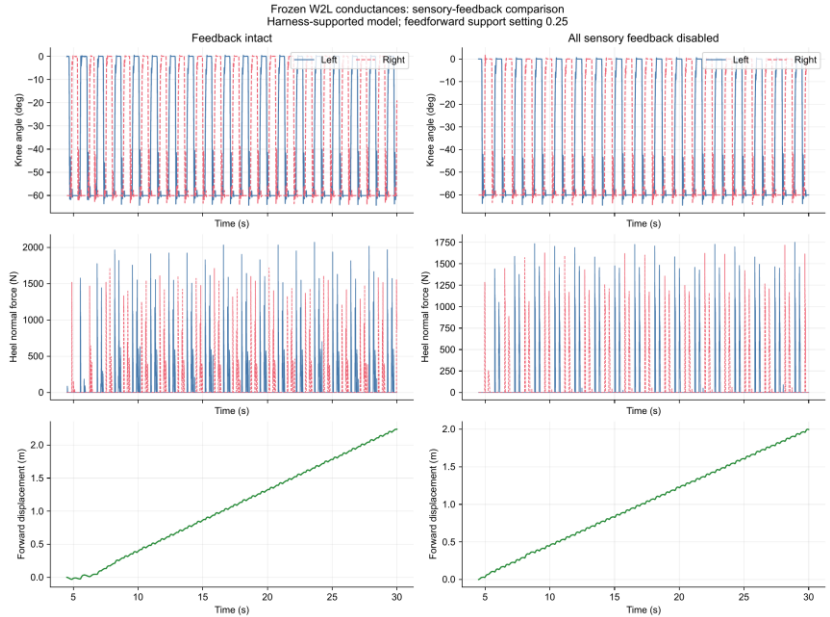
Stimulated layer	RG shift, cycles
RG	0.211
PF	0.000
MN	0.000

RG: rhythm generator
PF: pattern formation
MN: motor neuron

Reduced Deng-derived circuit · two baseline periods · suspended model

Dissertation Ch. 6 and Appendix D · Current saved verification records

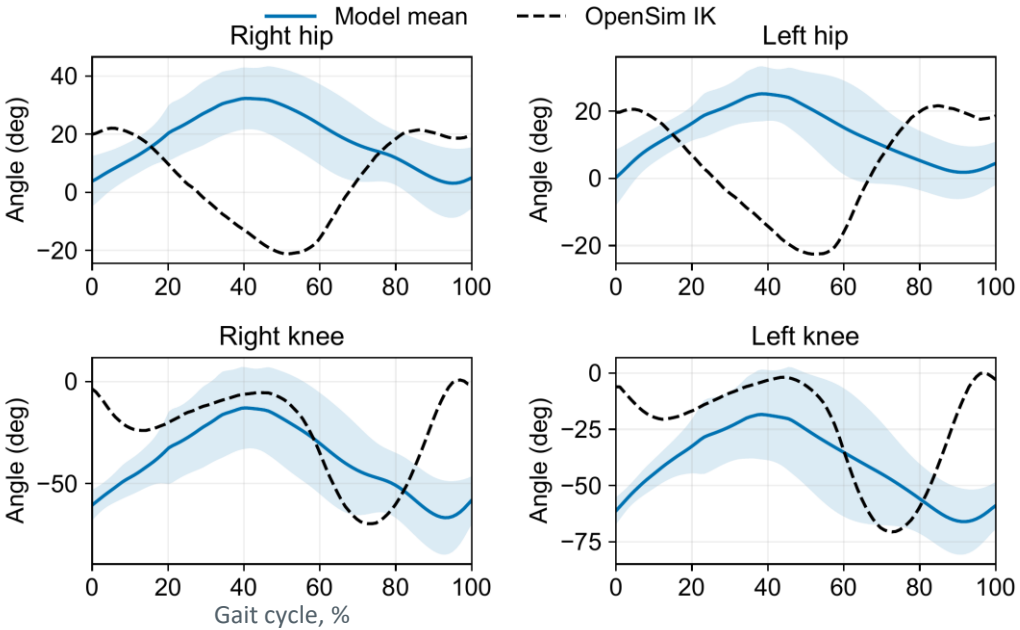
Feedback coordinates foot loading in the W2L model



Median bilateral contact phase: 0.498 intact · 0.223 without feedback

Dissertation Ch. 6 and Appendix D · Current saved verification records

Rhythmic Gait2392 cycles retain hip and knee errors



Hip timing differs

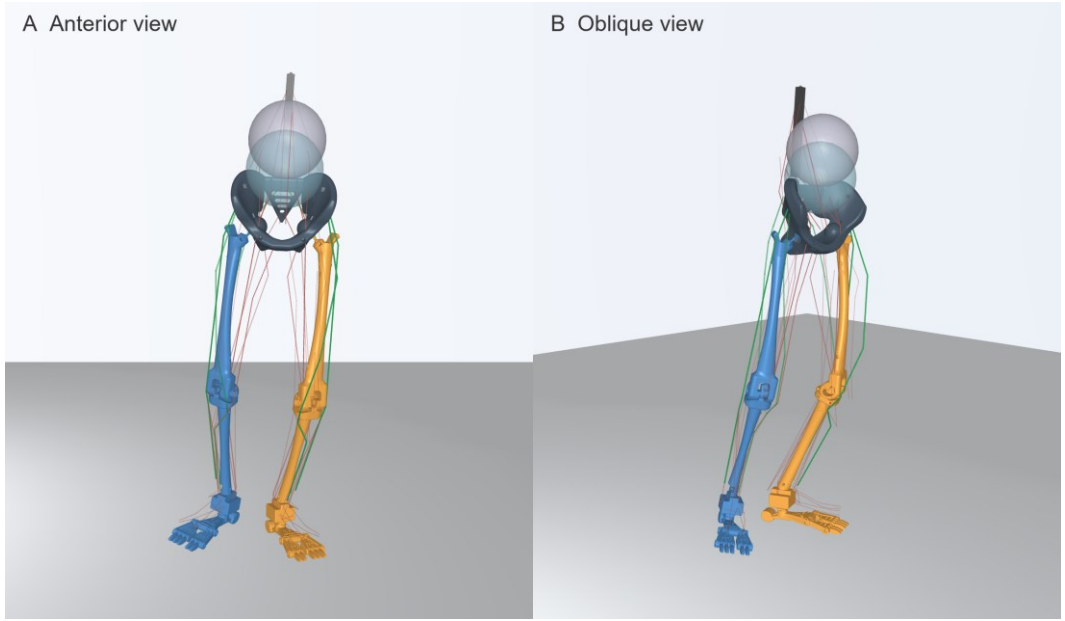
Knee timing differs

No fitted phase shift or amplitude scaling

10 cycles per leg · stance fractions 0.71 / 0.65 versus reference 0.61

Dissertation Ch. 6 and Appendix D · Current saved verification records

Extra abductor control prevents crossing in the AH001 test



15–40 s	Off	On
Crossed states	67%	0%
Min. spacing	−38.3 mm	142.4 mm

54 reference-tension actuators; 80% vertical + lateral/angular assistance
MN clamp cancels PF/reflex currents; regular stepping unverified

Dissertation Ch. 6 and Appendix D · Current saved verification records

These contributions enable better-defined research experiments

Contribution	Research enabled
BPA force model	Compare actuator lengths before building a route
Joint design factors and physical tests	Test transmission choices against torque requirements
SADb	Select feedback hypotheses with source context
SNS tools and controlled demonstrations	Distinguish neural, sensory, and mechanical effects

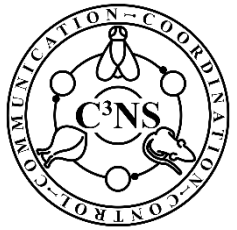
Future experiments can connect these foundations to the research goals

Research goal	Next experiments
Autonomy in unstructured environments	Dynamic loading, perturbations, and graded support reduction
Understanding biological mechanisms	Matched neural-pathway and mechanical interventions
Prosthetic and orthotic translation	Task-specific assistive concepts and later human validation

Greater biomimetic detail is useful when it enables a more decisive test

Dissertation Discussion; current model limitations

The dissertation makes key biomimetic design assumptions testable



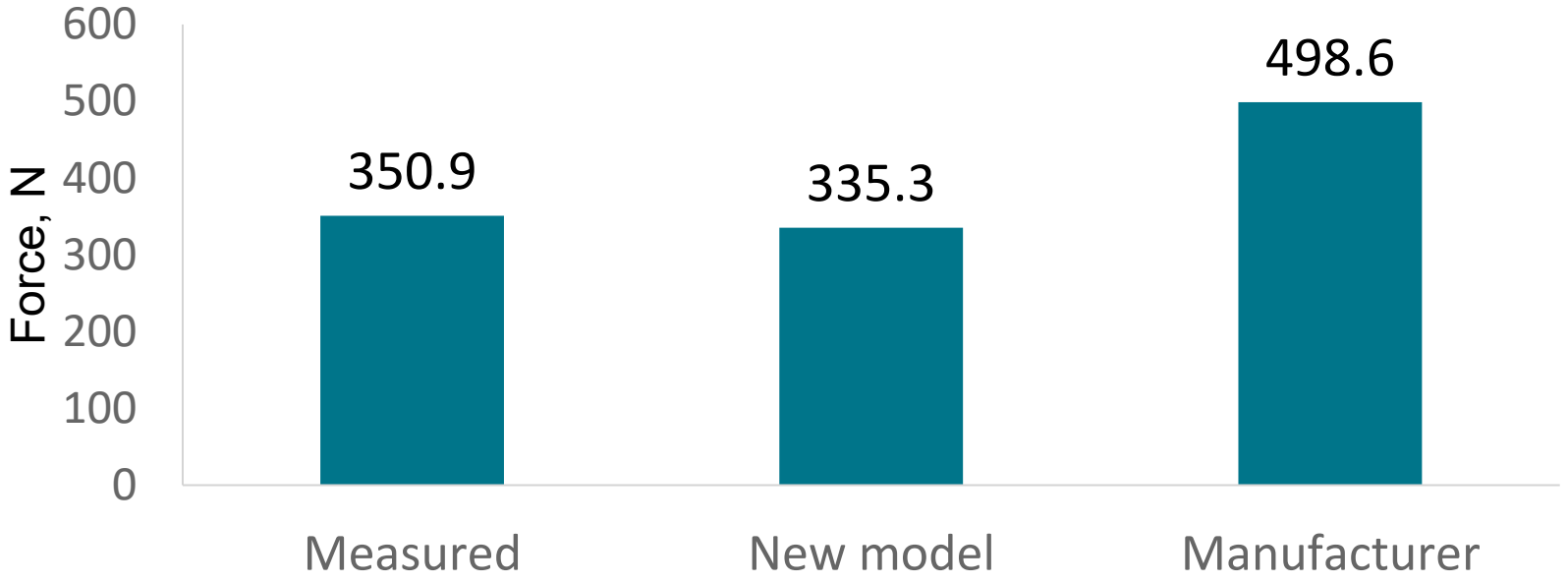
Actuator choice grounded in measured force

Joint designs tested against selected human targets

Sensory hypotheses made explicit and experimentally testable

Thank you · Questions

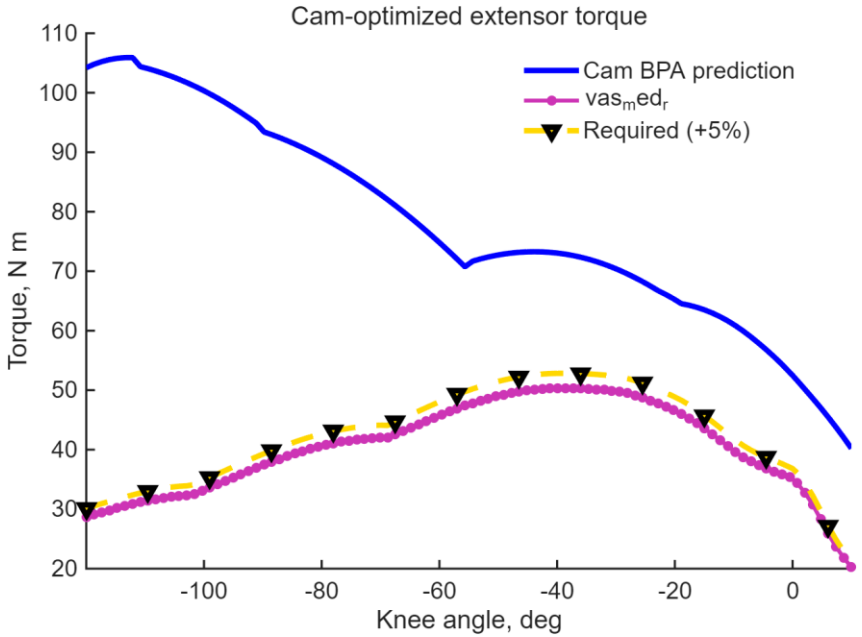
At 112 mm, the new model reduces force error to 4.5%



10 mm BPA · 112 mm resting length · 620 kPa

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

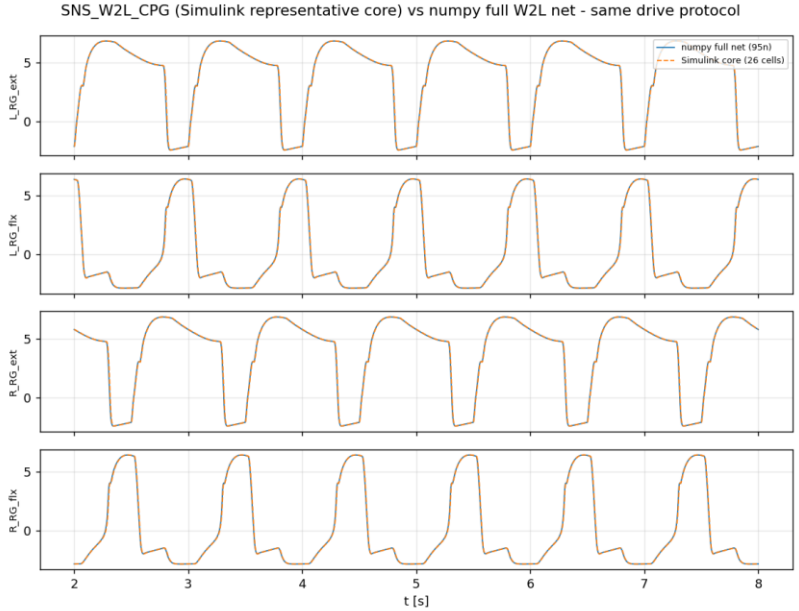
The cam model predicts headroom above the required target



+30.83% above required envelope · 269–507 kPa predicted pressure

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

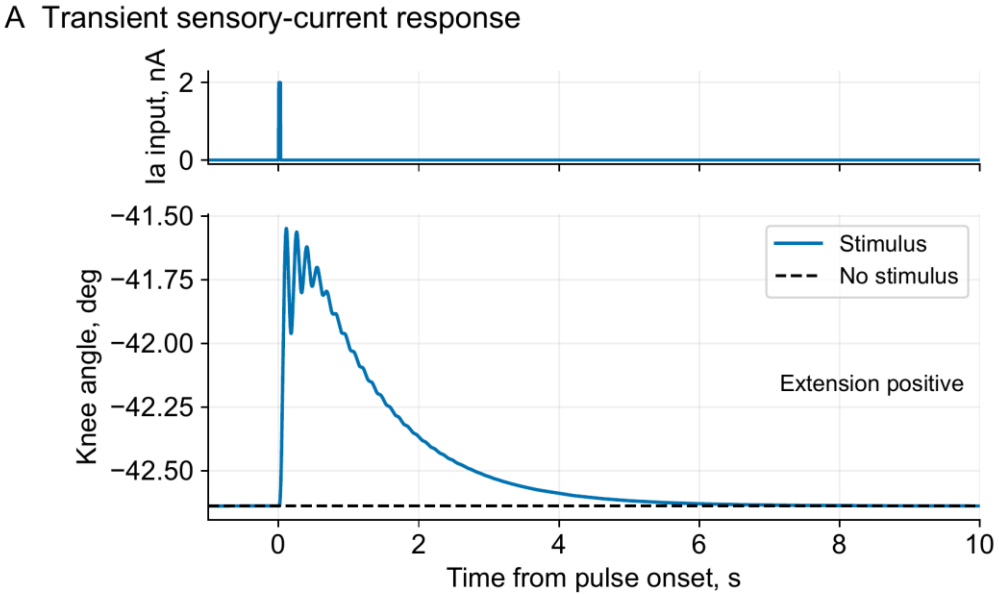
Matched rhythm cores reproduce the same driven bursts

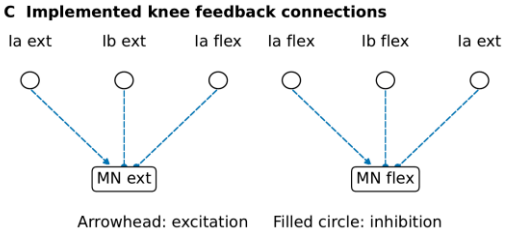


1.000 s period · 10 expected bursts · correlation ≥ 0.99996

Dissertation Ch. 6 and Appendix D · Current saved verification records

The knee model produces a transient sensory response





2 nA for 30 ms

1.09° at 111 ms

Recovery at 6.355 s

Six neural saturation widths changed in this configuration

Dissertation Ch. 6 and Appendix D · Current saved verification records

The evidence establishes specific capabilities within clear limits

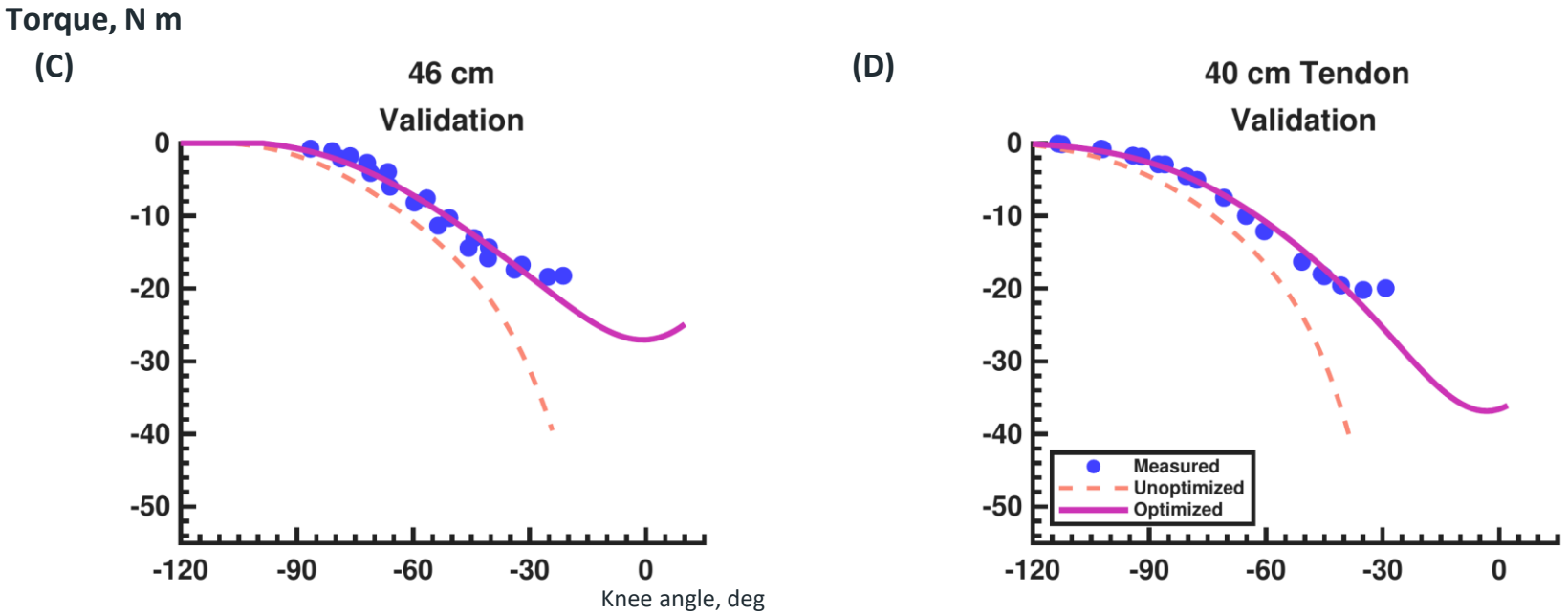
Evidence level	What the result establishes	Remaining boundary
Measured hardware	Isometric torque capacity	Dynamic response and transfer
Neural demonstrations	Specified response and intervention effect	Circuit- and protocol-specific
Supported W2L / human models	Coupling and coordination tests	External support remains
AH001 integration	Hybrid allocation; PF/reflex currents cancelled	80% vertical + lateral/angular support

AH001: reference tensions, MN clamp, provisional feet, and ideal brackets

No harness-free walking has been demonstrated

Current manuscript Ch. 6 / Appendix D · Session 1 AH001 records

The held-out flexor traces show the corrected torque trend



Held-out mean RMSE: 7.588 to 1.490 N m

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

The adopted identification reserves tests in each configuration

Configuration	Training	Held out	Not used
Pinned flexor	2	2	1
Pinned extensor	3	3	3

Shared effective stiffness pair: $\chi_1 = 39.98$, $\chi_2 = 14.73$ kN/m

Dissertation Ch. 4 · Current knee-torque manuscript, §§3–4

Physical dimensions differ from prediction settings

Actuator	Resting length, mm	Tendon length, mm
Flexor 1	561	28
Flexor 2	560	30
Extensor 1	828	92
Extensor 2	826	85

Prediction inputs: flexor 561.49 / 5.00 mm; extensor 827.74 / ≈92.49 mm

Torque manuscript, Physical Tests · Dissertation Appendix C dimensions

Stimulation duration changes the deletion interpretation

Protocol	RG shift, cycles	Motor shift, cycles
RG, two periods	0.211	0.211
PF, two periods	0.000	0.000
MN, two periods	0.000	0.000
PF, one second	0.000	0.208

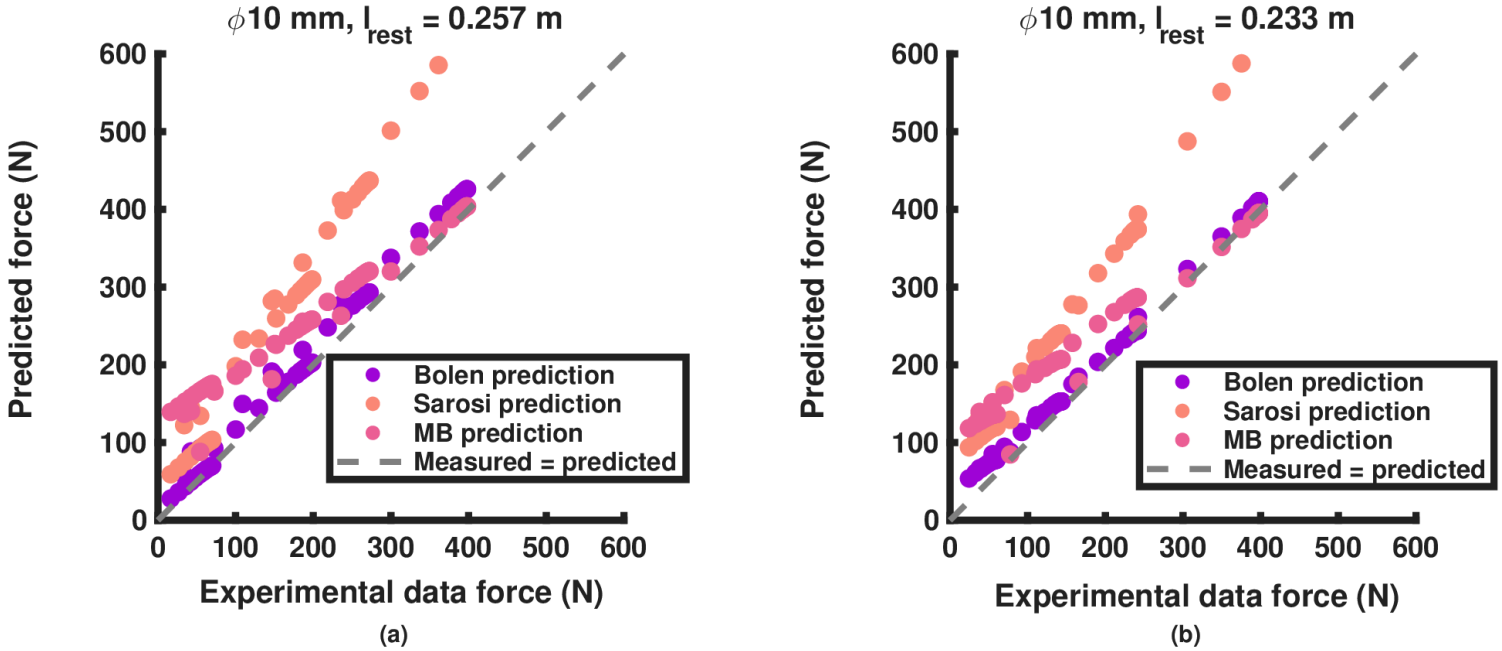
stimulus_results_all.csv · Current Appendix D layer-stimulus protocols

AH001 separation control changes a matched diagnostic

15–40 s measure	Control off	Control on
Reversed ankle ordering	67%	0%
Minimum separation	−38.3 mm	142.4 mm
Right TFL mean activation	22.3%	81.8%
Vertical assistance	80% BW	80% BW
Sampled opposite-leg penetration	2.700 mm	0 mm

AH001_20261028/PLACEMENT_RECOMMENDATION.md · asset_provenance.json

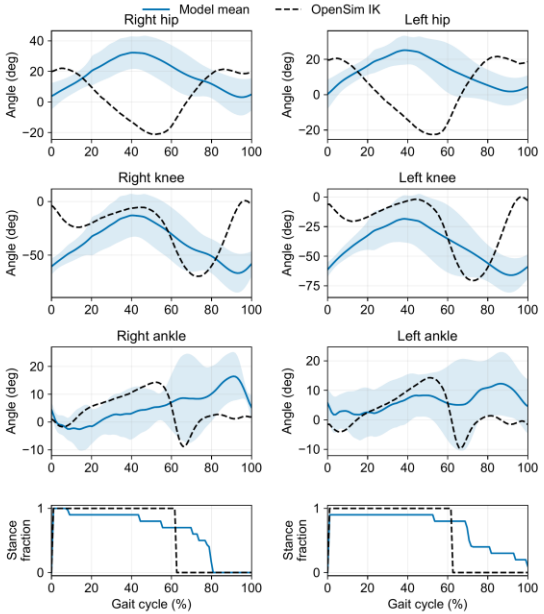
Reported force comparisons favor the model in the shown cases



Measured-versus-predicted force at the two displayed actuator lengths

Bolen et al. (2026), Actuators 15(5):230 · Dissertation Ch. 3

The complete IK overlay also exposes contact differences

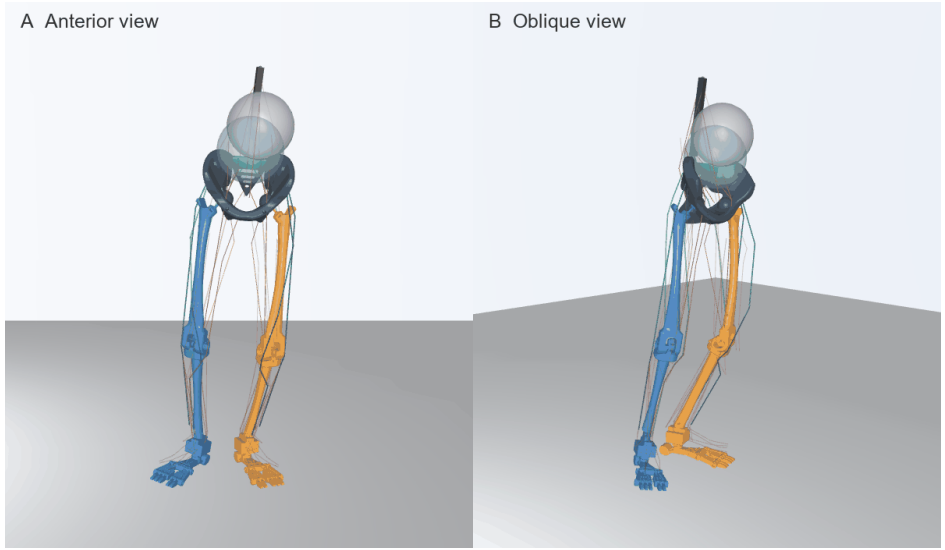


Model mean and standard deviation · OpenSim IK reference

Dissertation Ch. 6 and Appendix D · Current saved verification records

The saved AH001 replay shows supported hybrid motion

Supported CAD robot with increased abductor control



t = 20.00 s | vertical support 80% BW | minimum activation 5%

Hybrid control; reference tensions; lateral/angular assistance and knee-stop surrogates

Optional 12-second replay · same supported-control diagnostic

AH001_20261028/asset_provenance.json · saved 20–32 s interval